ABES ENGINEERING COLLEGE, GHAZIABAD

ECE DEPARTMENT

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Mini Project Report(BCC-351)

ON

Electricity Generation By Piezoelectric Sensor

Name Of Project : WALK WATT

Submitted by

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ABSTRACT

This project, "Walk Watt," explores the concept of harnessing human kinetic energy during walking to generate electricity. By integrating piezoelectric sensors into footwear, the mechanical stress exerted on the soles during each step is converted into electrical energy. This harvested energy can be utilized to power various small electronic devices, such as wearable fitness trackers, smartphones, or even contribute to powering ambient lighting in urban environments. The project aims to demonstrate the feasibility of piezoelectric energy harvesting as a sustainable and innovative approach to powering everyday electronics, minimizing reliance on traditional power sources and promoting a more self-sufficient and eco-friendlier lifestyle.

SOME KEY POINTS :

- Project Title: "Walk Watt"
- Core Concept: Harvesting energy from human walking by piezoelectric sensor
- Functionality: Converting mechanical energy (foot strikes) into electrical energy.
- Potential Applications: Powering wearable devices, contributing to urban energy.
- Project Goals: To show the feasibility and potential of piezoelectric energy .

INTRODUCTION

In an era of increasing energy demands and environmental concerns, the pursuit of sustainable and renewable energy sources has taken center stage. Traditional energy generation methods often rely on non-renewable resources, contributing to environmental degradation and climate change. However, the advent of energy harvesting technologies offers a promising alternative, enabling the conversion of ambient energy sources into usable electrical power. Among these technologies, piezoelectric energy harvesting has emerged as a particularly intriguing approach, capable of transforming mechanical energy into electrical energy.

The "Walk Watt" project capitalizes on this innovative technology by harnessing the kinetic energy generated during human walking. By integrating piezoelectric materials into footwear or insoles, the mechanical stress exerted on the soles during each step can be effectively converted into electrical energy. This harvested energy can then be utilized to power a range of electronic devices, from wearable fitness trackers and smartphones to ambient lighting systems in urban environments.

The concept of "Walk Watt" aligns with the growing trend of wearable technology and the increasing demand for portable power solutions. By seamlessly integrating energy harvesting capabilities into everyday activities, such as walking, this project aims to contribute to a more sustainable and self-sufficient energy landscape. Moreover, "Walk Watt" has the potential to revolutionize the way we power our devices, reducing our reliance on traditional power sources and minimizing our environmental impact.

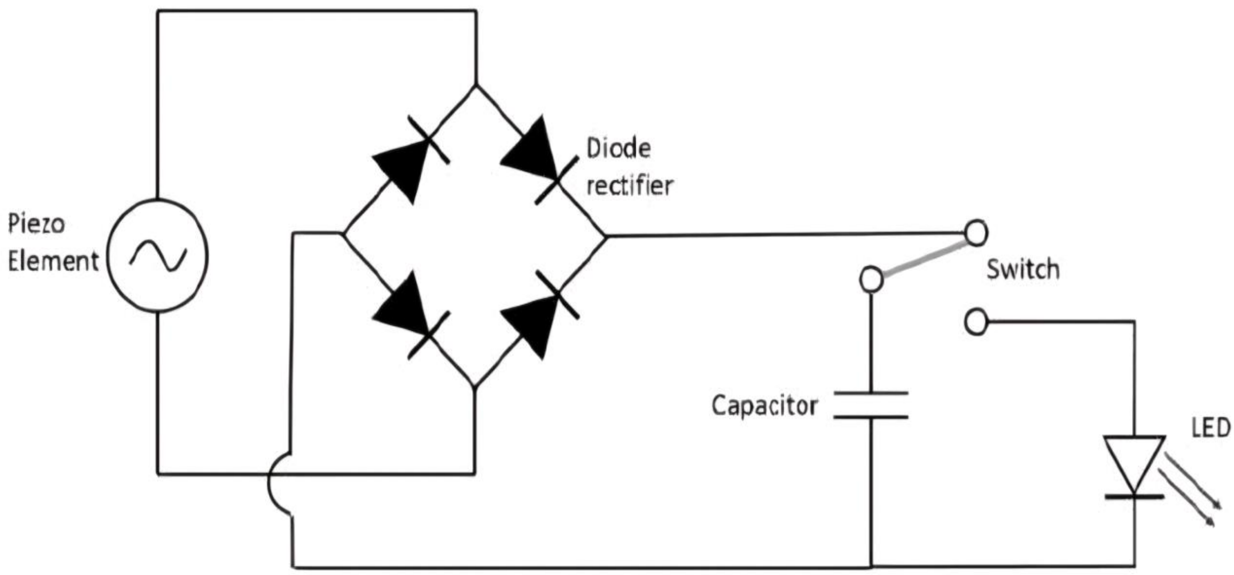
The Science Behind "Walk Watt": Piezoelectricity and Energy Conversion

At the heart of the "Walk Watt" project lies the fascinating phenomenon of piezoelectricity. Certain materials, known as piezoelectric materials, exhibit a unique property: they generate an electric charge when subjected to mechanical stress or pressure. This effect, known as the direct piezoelectric effect, forms the foundation of energy harvesting in "Walk Watt."

DESCRIPTION

"Walk Watt" is an innovative project that aims to harness the kinetic energy generated during human walking to produce electricity. By integrating piezoelectric sensors into footwear, the mechanical stress exerted on the soles during each step is converted into electrical energy. This harvested energy can then be utilized to power various electronic devices, such as wearable fitness trackers, smartphones, or even contribute to powering ambient lighting in urban environments.

Fig. Circuit Diagram Of Our Project



Key Features:

*Piezoelectric Sensor: Utilizes the piezoelectric effect to convert mechanical energy from foot strikes into electrical energy.

*Wearable Integration: Designed for seamless integration into footwear (e.g., insoles, shoe soles).

*Energy Harvesting: Captures and stores the harvested energy in a rechargeable battery or super capacitor.

*Power Output: Provides a regulated power output to power various electronic devices.

*Sustainability: Promotes a more sustainable and self-sufficient approach to powering electronic devices.